Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features:
- - Customer demographics (Age, Gender, Location, Subscription Status)
- - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
- - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

● Data Loading: Imported the dataset using **pandas**.

● Initial Exploration: Used **df.info()** to check structure and **df.describe()** for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes

Previous Purchases	Payment Method	Frequency of Purchases
14	Venmo	Fortnightly
2	Cash	Fortnightly
23	Credit Card	Weekly
49	PayPal	Weekly
31	PayPal	Annually

● Missing Data Handling: Checked for null values and imputed missing values in the **Review Rating** column using the median rating of each product category.

● Column Standardization: Renamed columns to snake case for better readability and documentation.

● Feature Engineering:

  ○ Created **age_group** column by binning customer ages.

  ○ Created **purchase_frequency_days** column from purchase data.

● Data Consistency Check: Verified if **discount_applied** and **promo_code_used** were redundant; dropped **promo_code_used**.

● Database Integration: Saved the clean DataFrame into a CSV file and then imported that CSV file into MySQL Workbench for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in MySQL to answer key business questions:

1. **Revenue by Gender –** Compared total revenue generated by male vs. female customers.

gender	total_revenue
Male	155438
Female	75191

2. **High-Spending Discount Users –** Identified customers who used discounts but still spent above the average purchase amount.

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88
24	94
32	79
33	67
35	91
37	69
40	60
41	76
43	100

And so on.

3. **Top 5 Products by Rating –** Found products with the highest average review ratings.

item_purchased	product_review_rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Skirt	3.79

4. **Shipping Type Comparison –** Compared average purchase amounts between Standard and Express shipping.

shipping_type	average_purchase_amount
Express	60.34
Standard	58.41

5. **Subscribers vs. Non-Subscribers –** Compared average spend and total revenue across subscription status.

subscription_status	total_customers	average_spend	total_revenue
Yes	1033	59.42	61377
No	2830	59.81	169252

6. **Discount-Dependent Products –** Identified 5 products with the highest percentage of discounted purchases.

item_purchased	discount_rate
Hat	49.67
Sneakers	49.66
Coat	48.43
Sweater	47.20
Pants	46.43

7. **Customer Segmentation –** Classified customers into New, Returning, and Loyal segments based on purchase history.

category	customer_count
Loyal	3085
Returning	695
New	83

8. **Top 3 Products per Category –** Listed the most purchased products within each category.

category	item_purchased	purchase_count	rn
Accessories	Jewelry	170	1
Accessories	Sunglasses	159	2
Accessories	Belt	159	3
Clothing	Blouse	169	1
Clothing	Shirt	168	2
Clothing	Pants	168	3
Footwear	Sandals	159	1
Footwear	Shoes	147	2
Footwear	Sneakers	145	3
Outerwear	Jacket	163	1

9. **Repeat Buyers & Subscriptions –** Checked whether customers with >5 purchases are more likely to subscribe.

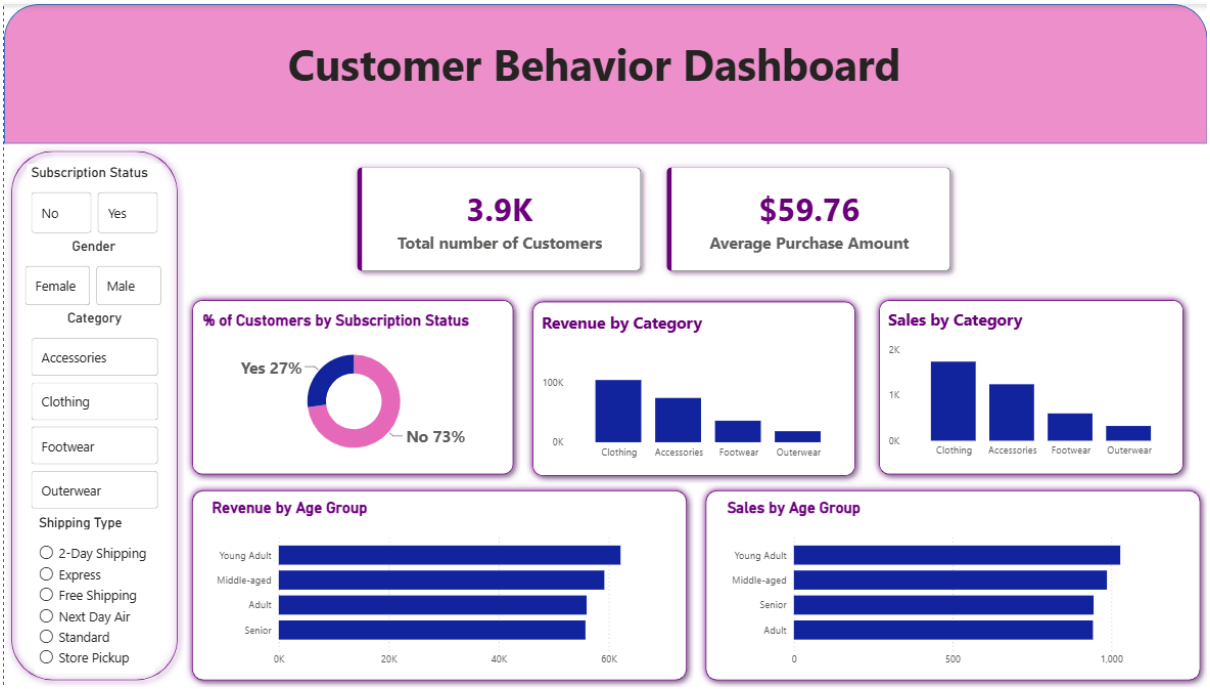
subscription_status	customer_count
Yes	939
No	2503

10. **Revenue by Age Group –** Calculated total revenue contribution of each age group.

age_group	total_revenue
Young Adult	61533
Middle-aged	58380
Adult	55569
Senior	55147

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



6. Business Recommendations

● Boost Subscriptions – Promote exclusive benefits for subscribers.

● Customer Loyalty Programs – Reward repeat buyers to move them into the “Loyal” segment.

● Review Discount Policy – Balance sales boosts with margin control.

● Product Positioning – Highlight top-rated and best-selling products in campaigns.

● Targeted Marketing – Focus efforts on high-revenue age groups and express-shipping users.